

# Package ‘tlsR’

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**Type** Package

**Title** Detection and Spatial Analysis of Tertiary Lymphoid Structures

**Version** 0.2.0

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**Description** Fast, reproducible detection and quantitative analysis of tertiary lymphoid structures (TLS) in multiplexed tissue imaging. Implements Independent Component Analysis Trace (ICAT) index, local Ripley's K scanning, automated K Nearest Neighbor (KNN)-based TLS detection, and T-cell clusters identification as described in Amiryousefi et al. (2025) <[doi:10.1101/2025.09.21.677465](https://doi.org/10.1101/2025.09.21.677465)>.

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**Depends** R (>= 4.0.0)

**Imports** dbscan (>= 1.1-10),  
fastICA (>= 1.2-3),  
FNN (>= 1.1.3),  
spatstat.explore (>= 3.0-0),  
spatstat.geom (>= 3.0-0),  
ggplot2 (>= 3.4.0),  
rlang (>= 1.0.0),  
methods

**Suggests** knitr,  
rmarkdown,  
testthat (>= 3.0.0)

**Encoding** UTF-8

**LazyData** true

**RoxygenNote** 7.3.3

**VignetteBuilder** knitr

**Config/testthat/edition** 3

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|              |                                                                             |
|--------------|-----------------------------------------------------------------------------|
| tlsR-package | <i>tlsR: Detection and Spatial Analysis of Tertiary Lymphoid Structures</i> |
|--------------|-----------------------------------------------------------------------------|

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**Description**

Fast, reproducible detection and quantitative analysis of tertiary lymphoid structures (TLS) in multiplexed tissue imaging data.

**Typical workflow**

1. Load or prepare a named list of data frames (ldata), one per tissue sample. Each data frame must contain columns x, y (spatial coordinates in microns), and phenotype (character: "B cell" / "T cell" / other).
2. Run [detect\\_TLS](#) to label B+T co-localised regions.
3. (Optional) Run [scan\\_clustering](#) to identify windows of significant immune clustering via local Ripley's L.
4. Run [calc\\_icat](#) to score the internal linearity/organisation of each detected TLS.
5. Run [detect\\_tic](#) to identify T-cell clusters outside TLS.
6. Use [summarize\\_TLS](#) to obtain a tidy summary table.
7. Use [plot\\_TLS](#) to produce publication-ready spatial plots.

**Author(s)**

**Maintainer:** Ali Amiryousefi <ali\_amiryousefi@hms.harvard.edu> ([ORCID](#))

Authors:

- Jeremiah Wala <jeremiah\_wala@dfci.harvard.edu> ([ORCID](#))

Other contributors:

- Peter Sorger ([ORCID](#)) [contributor]

**References**

Amiryousefi et al. (2025) [doi:10.1101/2025.09.21.677465](#)

**See Also**

Useful links:

- <https://github.com/amiryousefi/tlsR>
- Report bugs at <https://github.com/amiryousefi/tlsR/issues>

calc\_icat

*Calculate ICAT (Immune Cell Arrangement Trace) Index***Description**

Quantifies the linear / organised spatial arrangement of cells within a detected TLS by applying FastICA to the (x, y) coordinates of TLS cells and returning the trace of the estimated mixing matrix. Higher values indicate more structured (linear) cell organisation.

If the requested TLS contains fewer than 3 cells, or FastICA does not converge, the function returns NA\_real\_ with an informative message rather than throwing an error.

**Usage**

```
calc_icat(patientID, tlsID, ldata = NULL)
```

**Arguments**

|           |                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------|
| patientID | Character. Sample name in ldata.                                                                 |
| tlsID     | Numeric or integer. TLS identifier (value of tls_id_knn for the TLS of interest).                |
| ldata     | Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly). |

**Value**

A single numeric value (the ICAT index), or NA\_real\_ if computation is not possible (fewer than 3 cells, or FastICA did not converge).

**Examples**

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)
if (max(ldata[["ToySample"]]$tls_id_knn, na.rm = TRUE) > 0) {
  icat <- calc_icat("ToySample", tlsID = 1, ldata = ldata)
  icat
}
```

detect\_tic

*Detect Tumor-Infiltrating T-cell Clusters (TIC)***Description**

Applies HDBSCAN to T cells that lie outside of previously detected TLS regions to identify spatially compact T-cell clusters (TIC). Phenotype labels "T cell" and "T cells" are both accepted.

**Usage**

```
detect_tic(sample, min_pts = 10L, min_cluster_size = 10L, ldata = NULL)
```

**Arguments**

|                  |                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| sample           | Character. Sample name in ldata.                                                                                                            |
| min_pts          | Integer. HDBSCAN minPts parameter: minimum cluster size (default 10). Smaller values detect more, smaller clusters.                         |
| min_cluster_size | Integer. Minimum number of T cells for a HDBSCAN cluster to be retained; smaller clusters are merged back into noise (label 0). Default 10. |
| ldata            | Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly).                                            |

**Value**

The input ldata list with the sample data frame augmented by one new column:

tcell\_cluster\_hdbscan Integer. 0 = noise / not a T-cell cluster; positive integer = TIC cluster ID. Non-T-cell rows receive NA.

**Examples**

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)
ldata <- detect_tic("ToySample", ldata = ldata)
table(ldata[["ToySample"]]$tcell_cluster_hdbscan, useNA = "ifany")
```

---

detect\_TLS

*Detect Tertiary Lymphoid Structures using a KNN-density approach*

---

**Description**

Identifies TLS candidates by finding regions of high local B-cell density that also contain a sufficient number of nearby T cells (B+T co-localisation). Phenotype labels "B cell" and "B cells" (and their T-cell equivalents) are both accepted.

**Usage**

```
detect_TLS(
  LSP,
  k = 30L,
  bcell_density_threshold = 10,
  min_B_cells = 50L,
  min_T_cells_nearby = 20L,
  max_distance_T = 50,
  ldata = NULL
)
```

**Arguments**

|                         |                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|
| LSP                     | Character. Sample name in ldata.                                                                                    |
| k                       | Integer. Number of nearest neighbours used for density estimation (default 30, calibrated for 0.325 um/px imaging). |
| bcell_density_threshold | Numeric. Minimum average 1/k-distance (in microns) for a B cell to be considered locally dense (default 15).        |
| min_B_cells             | Integer. Minimum B cells per candidate TLS cluster (default 50).                                                    |
| min_T_cells_nearby      | Integer. Minimum T cells within max_distance_T microns of the candidate cluster centre (default 30).                |
| max_distance_T          | Numeric. Search radius (microns) for T-cell proximity check (default 50).                                           |
| ldata                   | Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly).                    |

**Value**

The input ldata list, with the data frame for LSP augmented by three new columns:

tls\_id\_knn Integer. 0 = non-TLS cell; positive integer = TLS cluster ID.

tls\_center\_x Numeric. X coordinate of the TLS centre for TLS cells; NA otherwise.

tls\_center\_y Numeric. Y coordinate of the TLS centre for TLS cells; NA otherwise.

**Examples**

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)
table(ldata[["ToySample"]][tls_id_knn])
plot(ldata[["ToySample"]][x, ldata[["ToySample"]][y,
      col = ifelse(ldata[["ToySample"]][tls_id_knn] > 0, "red", "gray"),
      pch = 19, cex = 0.5, main = "Detected TLS in toy data")
```

---

plot\_TLS

---

*Plot Spatial Map of TLS and T-cell Clusters*


---

**Description**

Produces a ggplot2 scatter plot of cell positions, coloured by TLS membership, T-cell cluster membership, and background phenotype.

**Usage**

```
plot_TLS(
  sample,
  ldata = NULL,
  show_tic = TRUE,
  point_size = 0.4,
  alpha = 0.6,
  tls_palette = c("#0072B2", "#009E73", "#CC79A7", "#D55E00", "#56B4E9", "#F0E442"),
  tic_colour = "#E69F00",
  bg_colour = "grey80"
)
```

**Arguments**

|             |                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------|
| sample      | Character. Sample name in ldata.                                                                 |
| ldata       | Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly). |
| show_tic    | Logical. Colour T-cell clusters in a distinct colour? Default TRUE.                              |
| point_size  | Numeric. Point size (default 0.4).                                                               |
| alpha       | Numeric. Point transparency (default 0.6).                                                       |
| tls_palette | Character vector of colours for TLS IDs. Recycled if there are more TLS than colours.            |
| tic_colour  | Character. Colour for T-cell cluster cells (default "#E69F00").                                  |
| bg_colour   | Character. Colour for background cells (default "grey80").                                       |

**Value**

A ggplot object (invisibly). Build and print the returned object to display the plot.

**Examples**

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)

p <- plot_TLS("ToySample", ldata = ldata)
print(p)
```

---

scan\_clustering

*Scan Tissue for Local Immune Cell Clustering (Ripley's L)*


---

**Description**

Applies a sliding-window centred L-function analysis across the tissue to identify spatially localised windows with statistically significant immune cell clustering. Each window is tested against a Monte Carlo envelope of complete spatial randomness (CSR).

**Usage**

```
scan_clustering(
  ws = 500,
  sample,
  phenotype = c("T cells", "B cells", "Both"),
  plot = FALSE,
  creep = 1L,
  nsim = 39L,
  min_cells = 10L,
  ldata = NULL
)
```

**Arguments**

|                        |                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ws</code>        | Numeric. Window side length in microns (default 500).                                                                                     |
| <code>sample</code>    | Character. Sample name in <code>ldata</code> .                                                                                            |
| <code>phenotype</code> | One of "T cells", "B cells", or "Both".                                                                                                   |
| <code>plot</code>      | Logical. Show a diagnostic plot for each significant window? (default FALSE).                                                             |
| <code>creep</code>     | Integer. Grid density factor; higher values give a finer scanning grid (default 1).                                                       |
| <code>nsim</code>      | Integer. Number of Monte Carlo simulations for envelope estimation (default 39, giving a pointwise significance level of 0.05 under CSR). |
| <code>min_cells</code> | Integer. Minimum cell count required in a window before it is analysed (default 10).                                                      |
| <code>ldata</code>     | Named list of data frames, or NULL to use the global <code>ldata</code> object (deprecated; pass explicitly).                             |

**Value**

A named list of results for significant windows. Each element is itself a list with:

`lest` The spatstat `lest` object.

`envelope` The Monte Carlo envelope object.

`window_center` Numeric vector `c(cx, cy)` of window centre coordinates.

`n_cells` Integer. Cell count in this window.

Returns an empty list (invisibly) when no significant windows are found.

**Examples**

```
data(toy_ldata)

models <- scan_clustering(ws = 500, sample = "ToySample",
                          phenotype = "B cells", plot = FALSE,
                          nsim = 19, ldata = toy_ldata)

length(models)
```

---

summarize\_TLS

*Summarize Detected TLS Across Samples*


---

**Description**

Produces a tidy data.frame with one row per sample summarising the number of detected TLS, their sizes, and (optionally) ICAT scores.

**Usage**

```
summarize_TLS(ldata, calc_icat_scores = FALSE)
```

**Arguments**

`ldata` Named list of data frames as returned by `detect_TLS` (and optionally `detect_tic`).  
`calc_icat_scores` Logical. Should ICAT scores be computed for each TLS and appended as a list-column? Default FALSE.

**Value**

A `data.frame` with columns:

- `sample` Sample name.
- `n_TLS` Number of TLS detected.
- `total_cells` Total cells in the sample.
- `TLS_cells` Number of cells assigned to any TLS.
- `TLS_fraction` Fraction of all cells that are TLS cells.
- `mean_TLS_size` Mean cells per TLS (NA if `n_TLS` = 0).
- `n_TIC` Number of T-cell clusters detected by `detect_tic` (NA if not yet run).
- `icat_scores` List-column of ICAT scores per TLS (only when `calc_icat_scores` = TRUE).

**Examples**

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)
summarize_TLS(ldata)
```

---

|                        |                                     |
|------------------------|-------------------------------------|
| <code>toy_ldata</code> | <i>Toy Multiplexed Imaging Data</i> |
|------------------------|-------------------------------------|

---

**Description**

A small synthetic dataset mimicking multiplexed tissue imaging data, used in package examples and tests. The list contains one sample named "ToySample".

**Usage**

```
toy_ldata
```

**Format**

A named list with one element:

`ToySample` A `data.frame` with the following columns:

- `x` Numeric. X coordinate in microns.
- `y` Numeric. Y coordinate in microns.
- `phenotype` Character. Cell phenotype label. Values are "B cell", "T cell", and "Other".

**Source**

Synthetically generated for package examples.

## References

Amiryousefi et al. (2025) [doi:10.1101/2025.09.21.677465](https://doi.org/10.1101/2025.09.21.677465)

## Examples

```
data(toy_ldata)
str(toy_ldata)
table(toy_ldata[["ToySample"]]$phenotype)
```